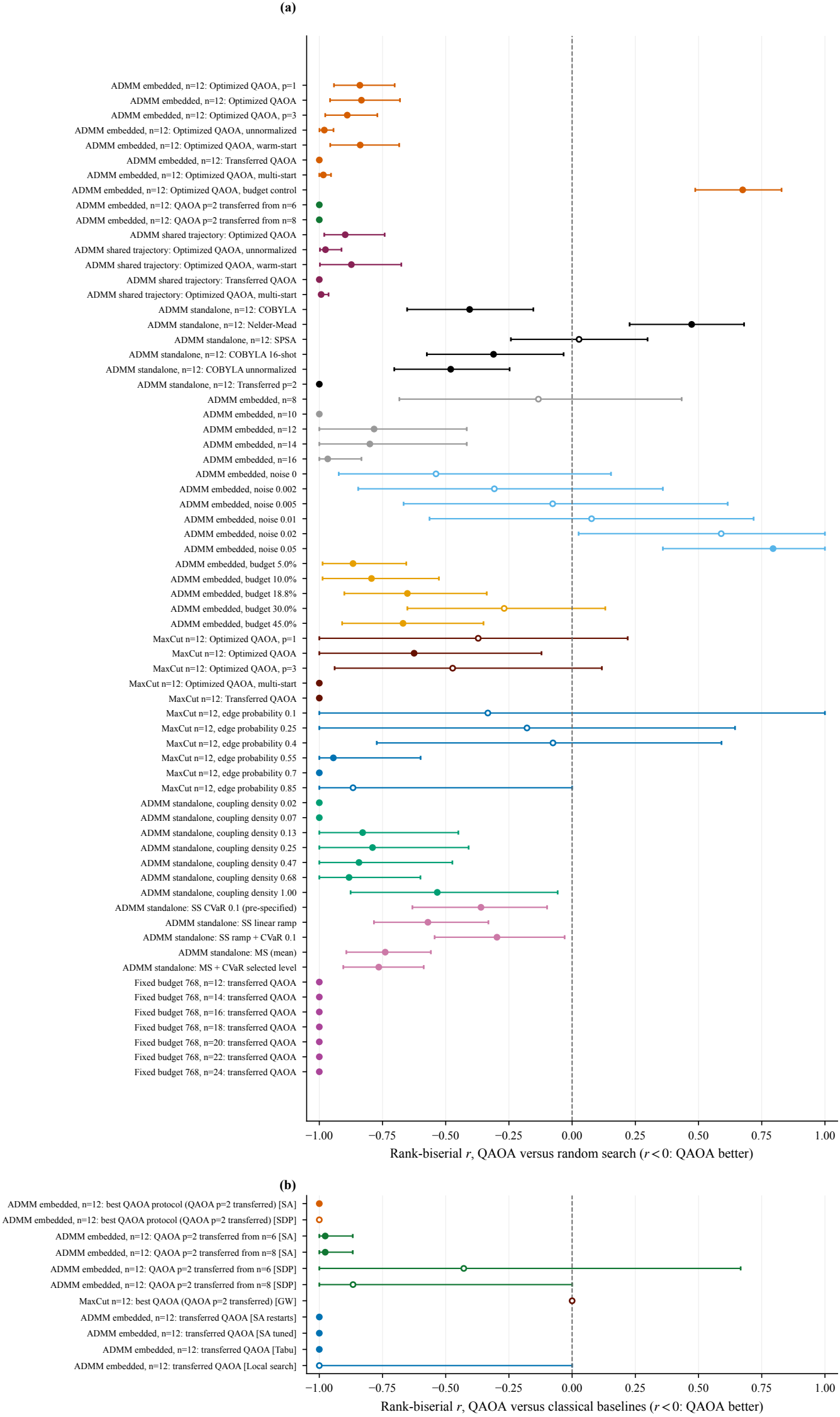

(a)
ADMM embedded, n=12: Optimized QAOA, p=1
ADMM embedded, n=12: Optimized QAOA
ADMM embedded, n=12: Optimized QAOA, p=3
ADMM embedded, n=12: Optimized QAOA, unnormalized
ADMM embedded, n=12: Optimized QAOA, warm-start
ADMM embedded, n=12: Transferred QAOA
ADMM embedded, n=12: Optimized QAOA, multi-start
ADMM embedded, n=12: Optimized QAOA, budget control
ADMM embedded, n=12: QAOA p=2 transferred from n=6
ADMM embedded, n=12: QAOA p=2 transferred from n=8
ADMM shared trajectory: Optimized QAOA
ADMM shared trajectory: Optimized QAOA, unnormalized
ADMM shared trajectory: Optimized QAOA, warm-start
ADMM shared trajectory: Transferred QAOA
ADMM shared trajectory: Optimized QAOA, multi-start
ADMM standalone, n=12: COBYLA
ADMM standalone, n=12: Nelder-Mead
ADMM standalone, n=12: SPSA
ADMM standalone, n=12: COBYLA 16-shot
ADMM standalone, n=12: COBYLA unnormalized
ADMM standalone, n=12: Transferred p=2
ADMM embedded, n=8
ADMM embedded, n=10
ADMM embedded, n=12
ADMM embedded, n=14
ADMM embedded, n=16
ADMM embedded, noise 0
ADMM embedded, noise 0.002
ADMM embedded, noise 0.005
ADMM embedded, noise 0.01
ADMM embedded, noise 0.02
ADMM embedded, noise 0.05
ADMM embedded, budget 5.0%
ADMM embedded, budget 10.0%
ADMM embedded, budget 18.8%
ADMM embedded, budget 30.0%
ADMM embedded, budget 45.0%
MaxCut n=12: Optimized QAOA, p=1
MaxCut n=12: Optimized QAOA
MaxCut n=12: Optimized QAOA, p=3
MaxCut n=12: Optimized QAOA, multi-start
MaxCut n=12: Transferred QAOA
MaxCut n=12, edge probability 0.1
MaxCut n=12, edge probability 0.25
MaxCut n=12, edge probability 0.4
MaxCut n=12, edge probability 0.55
MaxCut n=12, edge probability 0.7
MaxCut n=12, edge probability 0.85
ADMM standalone, coupling density 0.02
ADMM standalone, coupling density 0.07
ADMM standalone, coupling density 0.13
ADMM standalone, coupling density 0.25
ADMM standalone, coupling density 0.47
ADMM standalone, coupling density 0.68
ADMM standalone, coupling density 1.00
ADMM standalone: SS CVaR 0.1 (pre-specified)
ADMM standalone: SS linear ramp
ADMM standalone: SS ramp + CVaR 0.1
ADMM standalone: MS (mean)
ADMM standalone: MS + CVaR selected level
Fixed budget 768, n=12: transferred QAOA
Fixed budget 768, n=14: transferred QAOA
Fixed budget 768, n=16: transferred QAOA
Fixed budget 768, n=18: transferred QAOA
Fixed budget 768, n=20: transferred QAOA
Fixed budget 768, n=22: transferred QAOA
Fixed budget 768, n=24: transferred QAOA
−1.00
−0.75
−0.50
−0.25
0.00
0.25
0.50
0.75
1.00
Rank-biserial r, QAOA versus random search (r < 0: QAOA better)
(b)
ADMM embedded, n=12: best QAOA protocol (QAOA p=2 transferred) [SA]
ADMM embedded, n=12: best QAOA protocol (QAOA p=2 transferred) [SDP]
ADMM embedded, n=12: QAOA p=2 transferred from n=6 [SA]
ADMM embedded, n=12: QAOA p=2 transferred from n=8 [SA]
ADMM embedded, n=12: QAOA p=2 transferred from n=6 [SDP]
ADMM embedded, n=12: QAOA p=2 transferred from n=8 [SDP]
MaxCut n=12: best QAOA (QAOA p=2 transferred) [GW]
ADMM embedded, n=12: transferred QAOA [SA restarts]
ADMM embedded, n=12: transferred QAOA [SA tuned]
ADMM embedded, n=12: transferred QAOA [Tabu]
ADMM embedded, n=12: transferred QAOA [Local search]
−1.00
−0.75
−0.50
−0.25
0.00
0.25
0.50
0.75
1.00
Rank-biserial r, QAOA versus classical baselines (r < 0: QAOA better)